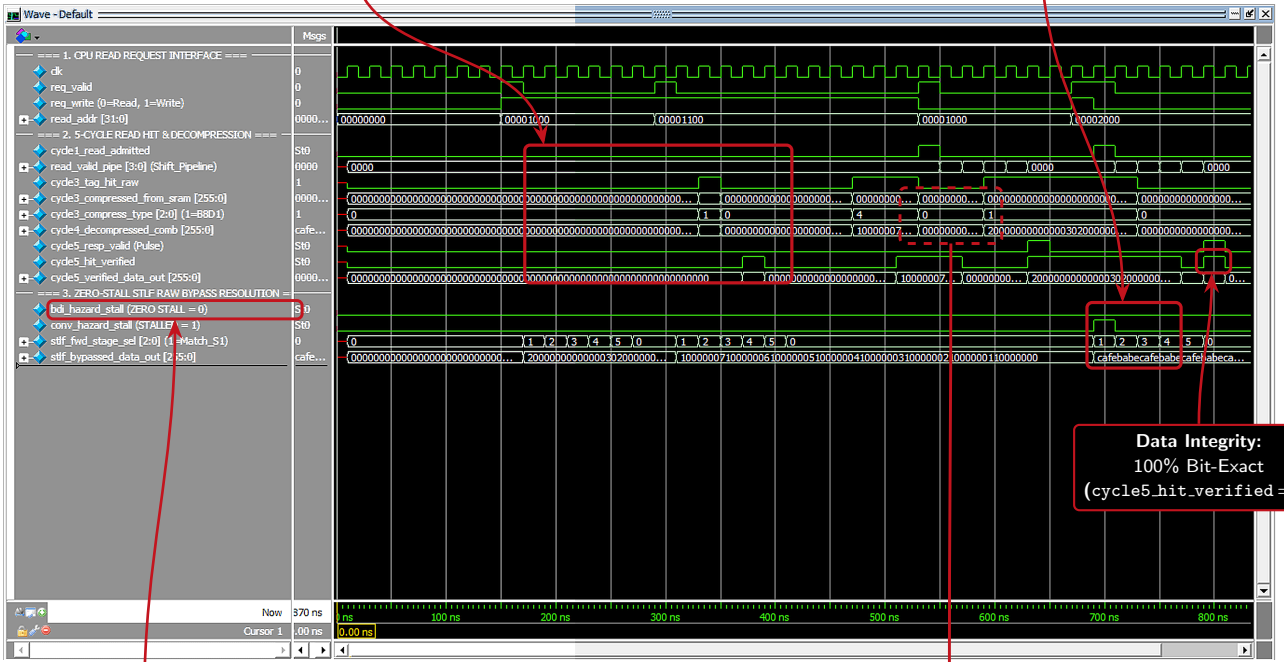
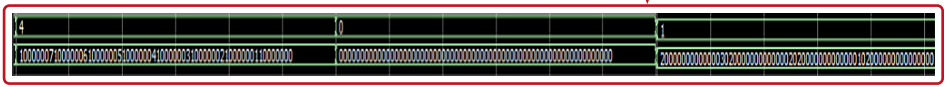
5-Cycle Read Hit Latency:
C1: Req → C2: Addr Route / Pipe Shift → C3: Tag & SRAM Read
C4: Comb Decompress → C5: Valid Strobe

Zero-Stall STLF Bypass ($t = 680$ ns):
RAW Hazard on Addr 0x2000
bdi_hazard_stall = 0 (0 Pipeline Stalls)



Data Integrity:
100% Bit-Exact
(cycle5_hit_verified = 1)

Hazard Interlock:
bdi_hazard_stall = 0
Zero Stalls Incurred



Magnified View: C3 Compressed Payload → C4 Restored Line